SIMATS ENGINEERING

Assessment Tool 3

Annexure A — Comparative Analysis

Course Name	Computer Networks
Course Code	CSA07
Course Outcome (CO4)	Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards (BL2, BL6)
Assessment Tool Weightage	30%
Total Marks	25

Code of Conduct

I ______________________________ (Name / Registration Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Introduction

A network topology defines the physical or logical arrangement in which devices, cables, and other networking components are interconnected to form a communication network. The choice of topology directly influences the cost, reliability, performance, scalability, and ease of maintenance of a network. The four most widely implemented topologies are Star, Bus, Mesh, and Hybrid, each of which is built using specific networking devices (hubs, switches, routers), transmission media (twisted pair, coaxial, fibre optic), and cabling standards (such as TIA/EIA-568).

This document presents a detailed comparative analysis of these four topologies across five perspectives — cost versus reliability and performance (Star vs Bus), fault tolerance (Mesh vs Star), installation complexity (Bus vs Mesh), scalability (Hybrid vs Star), and maintenance requirements (all four topologies). Each comparison is supported with conceptual diagrams, tabulated criteria, and a concluding insight to demonstrate a complete understanding of the design trade-offs involved in selecting an appropriate topology for a given networking scenario.

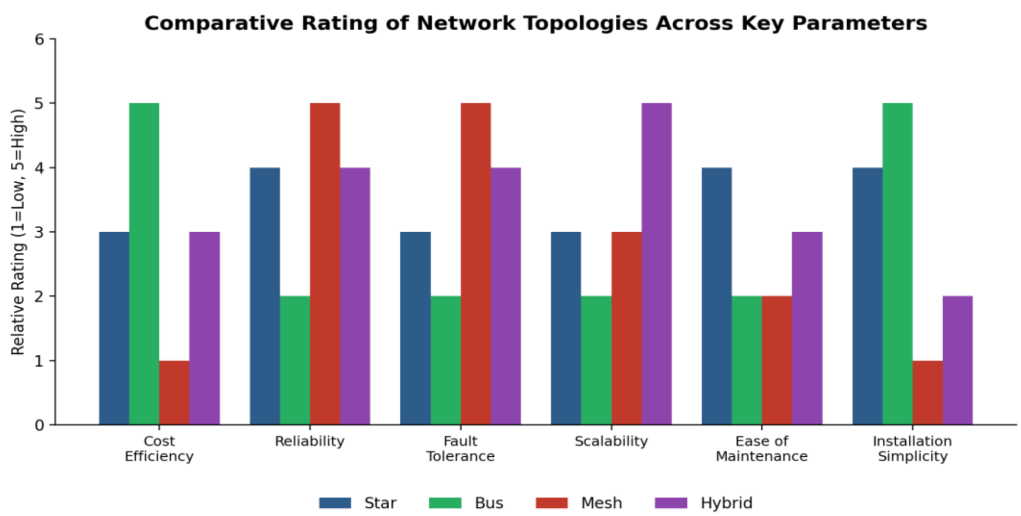

Figure 0: Overall comparative rating of Star, Bus, Mesh, and Hybrid topologies across six key parameters (1 = Low, 5 = High).

Question 1: Star Topology vs Bus Topology (Cost, Reliability, and Performance)

Overview

Star topology is a network arrangement in which every node (workstation, printer, or server) is connected individually to a central networking device such as a hub or a switch. All communication between nodes passes through this central device, which regulates traffic and forwards data frames to the intended recipient. Bus topology, in contrast, connects every node to a single shared communication line known as the backbone cable, terminated at both ends by terminators that prevent signal reflection.

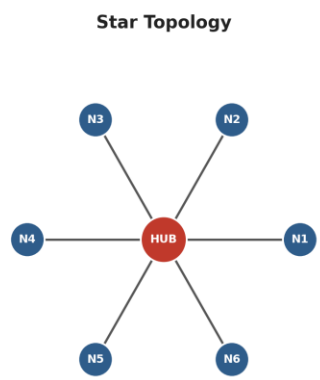

Figure 1(a): Star topology — every node (N1–N6) connects individually to a central hub/switch.

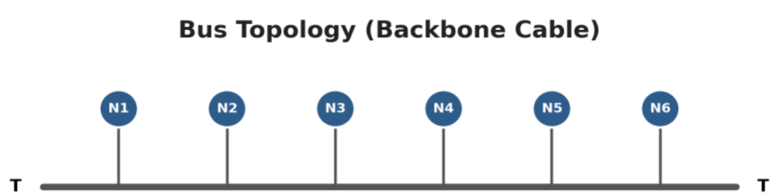

Figure 1(b): Bus topology — all nodes (N1–N6) share a single backbone cable terminated at both ends (T).

Cost Comparison

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub, and it consumes significantly more cabling since every node needs its own dedicated cable run back to the central point. In large installations with dozens of nodes spread across multiple rooms or floors, this cabling cost can become substantial, and the central switch itself represents an additional hardware investment. Bus topology, by comparison, uses a single backbone cable that all devices tap into, making it inherently low-cost in terms of both cabling material and installation labour. This is one of the reasons bus topology was popular in early small-office and home networks where budget was a primary constraint.

Reliability Comparison

Despite its higher cost, star topology is considerably more reliable than bus topology. Because each node has an independent cable connection to the central device, the failure of one node or one cable segment does not affect the operation of any other node — the rest of the network continues functioning normally. Bus topology does not offer this isolation: because every device shares the same backbone cable, a single break, loose connector, or faulty terminator anywhere along that cable can bring down communication for the entire network. This single point of failure along the backbone is widely regarded as the most significant weakness of bus topology.

Performance Comparison

Performance also favours star topology. Because a central switch manages and forwards traffic intelligently, collisions are minimised and each connected link can operate at close to its full available bandwidth. Bus topology, on the other hand, uses a shared medium in which only one device can transmit successfully at any given instant; as more devices are added, the probability of collisions increases and overall throughput degrades noticeably, especially under heavy load.

Parameter	Star Topology	Bus Topology
Cabling Cost	High — one cable per node	Low — single shared backbone
Central Device Cost	Required (hub/switch)	Not required
Reliability	High — node failure isolated	Low — backbone failure affects all
Performance	Better — managed by switch	Degrades as devices increase
Typical Use Case	Modern LANs, offices	Legacy small networks

Conclusion

In summary, the choice between star and bus topology represents a direct trade-off between upfront cost and long-term reliability/performance. Bus topology minimises installation expense but exposes the entire network to a single point of failure and degraded throughput as it scales. Star topology costs more to install but delivers superior fault isolation and consistently better performance, which is why it has become the dominant topology in almost all modern enterprise and home LAN deployments using structured cabling standards such as TIA/EIA-568.

Question 2: Mesh Topology vs Star Topology (Fault Tolerance)

Overview

Mesh topology is a network design in which every node maintains a direct point-to-point connection with multiple, and in a full mesh, every other node in the network. This creates numerous alternate paths through which data can travel from a source to a destination. Star topology, as described earlier, relies on a single central hub or switch through which all traffic must pass.

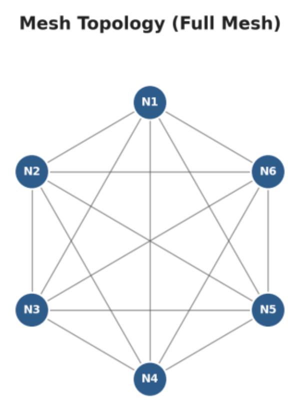

Figure 2: Mesh topology — every node (N1–N6) is directly connected to every other node, providing multiple alternate paths.

Fault Tolerance in Mesh Topology

Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing redundant paths for data transmission. If one link fails or becomes congested, data can automatically be rerouted through an alternate path without disrupting communication between the source and destination. This redundancy makes mesh topology extremely resilient, which is why it is commonly used in mission-critical environments such as military communication networks, backbone infrastructure of the internet, and wide area networks (WANs) connecting critical data centres.

Fault Tolerance in Star Topology

Star topology has only moderate fault tolerance. While the failure of an individual node's cable or network interface card only removes that one node from the network without affecting others, the central hub or switch represents a single point of failure for the entire network. If the central device stops functioning — due to power loss, hardware failure, or a firmware fault — every node connected to it loses network connectivity simultaneously, even though the individual node connections themselves remain physically intact.

Comparative Summary

Parameter	Mesh Topology	Star Topology
Redundant Paths	Multiple paths between any two nodes	Only one path, via central device
Single Point of Failure	None (in full mesh)	Central hub/switch
Impact of Node Failure	Negligible — rerouted automatically	Only that node is isolated
Impact of Central Device Failure	Not applicable	Entire network goes down
Suitability	Critical, high-availability systems	General-purpose LANs

Conclusion

Mesh topology is therefore more robust and reliable in critical environments compared to star topology, because its redundant interconnections eliminate any single point of failure. The trade-off is that this resilience comes at a significantly higher cost in cabling and network interfaces, which is why mesh topology is typically reserved for backbone or mission-critical links rather than for connecting ordinary end-user devices.

Question 3: Bus Topology vs Mesh Topology (Installation Complexity)

Overview

Installation complexity refers to the amount of planning, cabling, hardware ports, and labour required to physically set up a network topology. Bus and mesh topologies sit at opposite ends of this spectrum, making them a useful pair for comparison.

Installation of Bus Topology

Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it well suited for small networks with a limited number of devices confined to one physical area. Adding a new node typically requires only a single tap or connector onto the existing backbone cable, and no additional networking hardware such as switches is required. This simplicity is precisely why bus topology was widely adopted in early Ethernet installations using coaxial cable (10BASE2/10BASE5 standards).

Installation of Mesh Topology

Mesh topology is highly complex to install because each node must be directly connected to every other node, requiring a very large number of cables and network interface ports. For a full mesh network of n nodes, the number of required links is given by the formula $n(n-1)/2$. As shown in the table below, this number grows very quickly, which means that the complexity, cost, and time required to install a mesh network increases dramatically as the number of devices grows — making it time-consuming and costly to implement compared to the straightforward setup of bus topology.

Number of Nodes (n)	Bus: Cables Required	Mesh: Cables Required n(n-1)/2
4	1 backbone segment	6
6	1 backbone segment	15
8	1 backbone segment	28
10	1 backbone segment	45

Comparative Summary

Parameter	Bus Topology	Mesh Topology
Cabling Requirement	Minimal — one backbone	Very high — grows quadratically
Ports per Device	One (tap connector)	n − 1 ports per device
Installation Time	Short	Long, especially for large n
Planning Effort	Low	High — requires careful topology mapping
Best Suited For	Small, simple networks	Small critical cores, backbones

Conclusion

Bus topology's straightforward single-cable design makes it the easier and faster topology to install, particularly for small networks, whereas mesh topology's requirement for direct connections between every pair of nodes makes installation increasingly impractical as the network grows in size. Consequently, mesh topology is rarely deployed at full scale for end-user connectivity and is instead used selectively — for example, only among core routers or critical servers — to balance resilience against installation cost.

Question 4: Hybrid Topology vs Star Topology (Scalability)

Overview

Hybrid topology is a network design that combines two or more different basic topologies — such as star, mesh, or bus — into a single unified network, typically by linking multiple star or mesh segments together through backbone switches or routers. Scalability refers to how easily a network can grow to accommodate additional devices or segments without requiring a fundamental redesign.

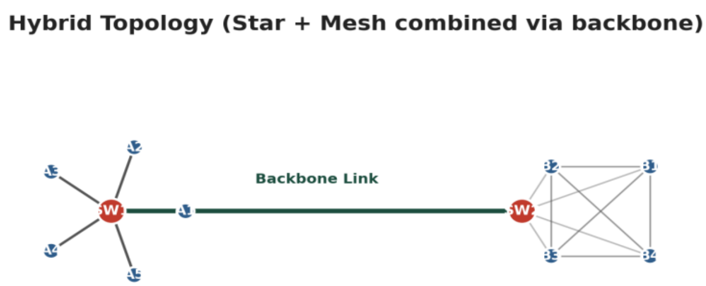

Figure 3: Hybrid topology — a star segment (SW1) and a mesh segment (SW2) joined through a backbone link, illustrating flexible expansion.

Scalability of Hybrid Topology

Hybrid topology is highly scalable because it combines two or more different topologies, allowing the network to expand flexibly according to organisational needs. For example, a new department can be added as an entirely new star segment and simply connected to the existing backbone, without disturbing the wiring or configuration of any other segment. This modular approach means new segments can be added without affecting the existing network structure, making hybrid topology the preferred choice for large campus networks, multi-building enterprises, and internet service provider networks.

Scalability of Star Topology

Star topology is also scalable, but only to a limited extent, because its growth is directly constrained by the port capacity of the central hub or switch. Once the central device reaches its port limit, expanding the network requires purchasing and installing additional switches, which then need to be interconnected — effectively pushing the network toward a hybrid design in any case. Star topology alone, therefore, does not scale gracefully beyond the capacity of a single central device or a small stack of devices.

Comparative Summary

Parameter	Hybrid Topology	Star Topology
Expansion Method	Add new segments via backbone	Limited by central device ports
Impact on Existing Network	Minimal — modular addition	May require re-cabling / new switch
Suitable Scale	Large campus / enterprise networks	Small to medium single-location LANs
Flexibility	Very high — mixes topology types	Moderate — single topology only
Long-term Cost of Growth	Predictable, incremental	Can spike once port limit reached

Conclusion

Hybrid topology provides greater scalability than star topology because it is not bound to the limitations of a single central device; instead, it grows by federating multiple star, bus, or mesh segments through backbone connections. This makes hybrid topology the standard architectural approach for large, evolving organisations, while pure star topology remains best suited to smaller, relatively static networks confined to a single location.

Question 5: Maintenance Comparison Across Star, Mesh, Bus, and Hybrid Topologies

Overview

Maintenance complexity refers to the effort required to monitor a network, detect faults, isolate problems, and carry out repairs or upgrades without causing extended downtime. This final comparison considers all four topologies together, since maintenance is where their structural differences become most apparent in day-to-day network administration.

Star Topology Maintenance

Star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable — network administrators can check the status LEDs on the central switch for each port to instantly identify which link has failed, without disrupting any other connected device.

Mesh Topology Maintenance

Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and the large number of interconnections between nodes, making troubleshooting a time-consuming process, since a fault could potentially be located along any one of many possible paths between two communicating devices.

Bus Topology Maintenance

Bus topology is harder to maintain than star topology because faults occurring anywhere in the shared backbone cable can disrupt the entire network and are often difficult to physically locate, sometimes requiring specialised cable testers to pinpoint the exact break or fault along the cable run.

Hybrid Topology Maintenance

Hybrid topology has moderate maintenance complexity; while it benefits from the flexibility of combining multiple topology types, managing the different combined segments requires careful monitoring and skilled network administration to ensure that faults in one segment are correctly isolated and do not cascade into the backbone connecting the other segments.

Consolidated Maintenance Comparison

Topology	Fault Isolation	Troubleshooting Effort	Overall Maintenance Level
Star	Easy — per-port LED status	Low	Easy
Mesh	Difficult — many possible paths	High	Difficult
Bus	Difficult — fault anywhere on backbone	Medium–High	Difficult
Hybrid	Moderate — depends on segment	Medium	Moderate

Conclusion

Across all four topologies, star topology offers the simplest maintenance experience due to its centralised monitoring point, while mesh and bus topologies demand considerably more administrative effort because faults are harder to isolate. Hybrid topology occupies a middle ground: it inherits some of the monitoring convenience of its star segments while requiring skilled oversight to manage the interaction between its combined sub-topologies.